

Probability Foundations

Technical reference

Companion to the note on lorenzoliuzzo.github.io. Axioms, expectation, variance and limit theorems, with proofs.

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1 Probability toolkit

1.1 Probability spaces and the axioms

A probability space is a triple $(\Omega, \mathcal{F}, P)$: a sample space Ω , a σ -algebra $\mathcal{F}$ of events (a family of subsets of Ω containing Ω and closed under complements and countable unions), and a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the axioms below. Events A, B are *disjoint* if $A \cap B = \emptyset$; events $B_1, B_2, \dots$ form a *partition* if they are pairwise disjoint and $\bigcup_i B_i = \Omega$.

Definition (Kolmogorov axioms). (A1) $P(A) \geq 0$ for every event A . (A2) $P(\Omega) = 1$. (A3) For pairwise disjoint events $A_1, A_2, \dots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem (Consequences of the axioms). For events $A, B, A_1, A_2, \dots$: (i) $P(\emptyset) = 0$; (ii) finite additivity: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for disjoint A_i ; (iii) $P(A^c) = 1 - P(A)$; (iv) if $A \subseteq B$ then $P(A) \leq P(B)$; in particular $P(A) \leq 1$; (v) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; (vi) union bound: $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (i) Take $A_i = \emptyset$ for all i in (A3): $P(\emptyset) = \sum_i P(\emptyset)$, which forces $P(\emptyset) = 0$. (ii) Apply (A3) with $A_i = \emptyset$ for $i > n$ and use (i). (iii) $\Omega = A \cup A^c$ is a disjoint union, so $1 = P(A) + P(A^c)$. (iv) $B = A \cup (B \setminus A)$ is disjoint, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$; with $B = \Omega$ this gives $P(A) \leq 1$. (v) Write $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$, both disjoint; then $P(B \setminus A) = P(B) - P(A \cap B)$, and substituting gives (v). (vi) Set $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j < i} A_j$. The B_i are disjoint, $\bigcup_i B_i = \bigcup_i A_i$ and $B_i \subseteq A_i$, so by (A3) and (iv) $P(\bigcup_i A_i) = \sum_i P(B_i) \leq \sum_i P(A_i)$. $\square$

The general inclusion-exclusion formula is $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots$. Two further facts follow from (A3) by taking limits: if A_n increases to A then $P(A_n) \rightarrow P(A)$, and likewise for decreasing sequences (continuity of P). When Ω is finite and outcomes are equally likely, $P(A) = |A| / |\Omega|$.

1.2 Conditional probability, independence, Bayes

Definition. For $P(B) > 0$, the conditional probability of A given B is $P(A | B) = P(A \cap B) / P(B)$. Then $P(\cdot | B)$ is itself a probability: it satisfies (A1)–(A3) on $\mathcal{F}$, so all the consequences above hold conditionally.

Rearranging gives the multiplication rule $P(A \cap B) = P(A | B) P(B)$ and, by induction, the chain rule

$$P(A_1 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) \dots P(A_n | A_1 \cap \dots \cap A_{n-1}). \quad (1)$$

Definition (Independence). A and B are independent if $P(A \cap B) = P(A) P(B)$ (equivalently $P(A | B) = P(A)$ when $P(B) > 0$). Events $A_1, \dots, A_n$ are *mutually independent* if $P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i)$ for every subset S of indices; pairwise independence alone does not imply this. A and B are conditionally independent given C if $P(A \cap B | C) = P(A | C) P(B | C)$.

Remark. Disjoint events with positive probability are never independent: $P(A \cap B) = 0 \neq P(A) P(B)$.

Theorem (Law of total probability and Bayes' theorem). Let $B_1, B_2, \dots$ be a partition of Ω with $P(B_i) > 0$. For any event A ,

$$P(A) = \sum_i P(A | B_i) P(B_i), \quad (2)$$

and, if $P(A) > 0$,

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_i P(A | B_i) P(B_i)}. \quad (3)$$

Proof. The sets $A \cap B_i$ are disjoint with union A , so by (A3) and the multiplication rule $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i) P(B_i)$. For Bayes, $P(B_j | A) = P(A \cap B_j) / P(A) = P(A | B_j) P(B_j) / P(A)$, and the denominator is Equation 2. $\square$

In Bayesian language $P(B_j)$ is the prior, $P(A | B_j)$ the likelihood, $P(A)$ the evidence (marginal likelihood) and $P(B_j | A)$ the posterior. The odds form is $P(B | A) / P(B^c | A) = [P(A | B) / P(A | B^c)] \cdot [P(B) / P(B^c)]$: posterior odds equal likelihood ratio times prior odds.

Example (Diagnostic test). Prevalence $P(D) = 0.01$, sensitivity $P(+ | D) = 0.95$, false-positive rate $P(+ | D^c) = 0.10$. Then

$$P(D | +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.088. \quad (4)$$

The posterior is below 9% because the healthy majority produces more false positives than the sick minority produces true ones: the base rate matters.

1.3 Random variables and distributions

A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$. Its distribution function is $F(x) = P(X \leq x)$.

Lemma (Properties of the distribution function). F is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. For $a < b$, $P(a < X \leq b) = F(b) - F(a)$.

Proof. Monotonicity is (iv) applied to $\{X \leq x\} \subseteq \{X \leq y\}$. Right-continuity and the limits follow from continuity of P along monotone sequences of events. Finally $\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$ is disjoint. $\square$

X is discrete with mass function $p(x) = P(X = x)$, $\sum_x p(x) = 1$, or continuous with density $f = F'$, $\int f = 1$ and $P(a \leq X \leq b) = \int_a^b f$; in the continuous case $P(X = x) = 0$ for every x . The quantile function is $F^{-1}(p) = \inf\{x : F(x) \geq p\}$ and the median is $F^{-1}(1/2)$. For a pair (X, Y) with joint density $f_{X,Y}$, the marginal is $f_{X(x)} = \int f_{X,Y}(x, y) dy$, the conditional density is $f_{X|Y}(x | y) = f_{X,Y}(x, y) / f_{Y(y)}$, and X, Y are independent iff $f_{X,Y} = f_X f_Y$.

1.4 Expectation

Definition. $E[X] = \sum_x xp(x)$ (discrete) or $\int xf(x) dx$ (continuous), when the sum or integral converges absolutely. For a function g (law of the unconscious statistician), $E[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x) dx$.

Theorem (Properties of expectation). (i) Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$ for any X, Y , dependent or not. (ii) Monotonicity: $X \leq Y$ implies $E[X] \leq E[Y]$. (iii) If X, Y are independent, $E[XY] = E[X]E[Y]$. (iv) $E[\mathbb{1}_A] = P(A)$. (v) Jensen: for convex g , $g(E[X]) \leq E[g(X)]$.

Proof. (i) and (ii) follow from linearity and monotonicity of sums and integrals, using the joint density for X and Y together. (iii) By independence $f_{X,Y} = f_X f_Y$, so $E[XY] = \int \int xy f_X(x) f_Y(y) dx dy = E[X]E[Y]$. (iv) $\mathbb{1}_A$ takes the value 1 with probability $P(A)$ and 0 otherwise. (v) A convex g lies above its tangent line at $m = E[X]$: $g(x) \geq g(m) + c(x - m)$ for some slope c ; take expectations. $\square$

1.5 Variance, covariance, correlation

Definition. $\text{Var}(X) = E[(X - \mu)^2]$ with $\mu = E[X]$; the standard deviation is $\sigma = \sqrt{\text{Var}(X)}$. $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$ and, when $\sigma_X, \sigma_Y > 0$, the correlation is $\rho_{XY} = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$. Moments: $E[X^k]$; central moments $E[(X - \mu)^k]$; skewness $E[(X - \mu)^3]/\sigma^3$ and kurtosis $E[(X - \mu)^4]/\sigma^4$ (equal to 3 for the normal).

Theorem (Properties of variance and covariance). (i) $\text{Var}(X) = E[X^2] - (E[X])^2 \geq 0$, with equality iff X is almost surely constant. (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$. (iii) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$, $\text{Cov}(X, X) = \text{Var}(X)$, and Cov is bilinear. (iv) $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. (v) Independent X, Y have $\text{Cov}(X, Y) = 0$; the converse is false. (vi) $|\rho_{XY}| \leq 1$, with equality iff $Y = aX + b$ almost surely.

Proof. (i) Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations; nonnegativity holds because the integrand is ≥ 0 , and $E[(X - \mu)^2] = 0$ forces $X = \mu$ almost surely. (ii) $aX + b - E[aX + b] = a(X - \mu)$. (iii) Expand the product and use linearity; bilinearity is linearity in each argument. (iv) is (ii) and (iii) combined. (v) follows from (iii) and property (iii) of expectation. For the failure of the converse, let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$: $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$ since odd moments of a symmetric law vanish, yet Y is a function of X . (vi) For every real t , $0 \leq \text{Var}(Y - tX) = \text{Var}(Y) - 2t \text{Cov}(X, Y) + t^2 \text{Var}(X)$. A quadratic in t that never goes negative has non-positive discriminant, $4\text{Cov}(X, Y)^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0$, which is $|\rho| \leq 1$. Equality means the discriminant vanishes, so $\text{Var}(Y - tX) = 0$ for some t and $Y - tX$ is constant. $\square$

Property (iv) generalizes to $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, and so, for uncorrelated (hence independent) X_i with common variance σ^2 ,

$$\text{Var}\left(\sum_i X_i\right) = n\sigma^2, \quad \text{Var}(\bar{X}_n) = \sigma^2/n. \quad (5)$$

Theorem (Law of total expectation and total variance). If $E[X^2] < \infty$, then $E[X] = E[E[X | Y]]$ and $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$. $\square$

Proof. The first identity is the law of total probability for expectations: $E[X] = \int E[X | Y = y] f_Y(y) dy$. For the second, write $X - E[X] = (X - E[X | Y]) + (E[X | Y] - E[X])$ and square. The cross term has expectation zero: conditionally on Y the second factor is fixed and $E[X - E[X | Y] | Y] = 0$. The first square has expectation $E[\text{Var}(X | Y)]$ and the second $\text{Var}(E[X | Y])$. $\square$

1.6 Markov and Chebyshev

Lemma (Markov and Chebyshev). If $X \geq 0$ and $a > 0$ then $P(X \geq a) \leq E[X]/a$. For any X with mean μ and variance σ^2 and any $k > 0$, $P(|X - \mu| \geq k) \leq \sigma^2/k^2$.

Proof. $a \mathbb{1}\{X \geq a\} \leq X$ pointwise, take expectations. Chebyshev is Markov applied to $(X - \mu)^2$ and $a = k^2$. $\square$

The standard discrete and continuous distributions, with their means, variances and uses, are in the Sampling Distributions reference.

1.7 Limit theorems

Write $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

Theorem (Weak law of large numbers). If X_i are i.i.d. with $E X_1 = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$, then $\bar{X}_n \xrightarrow{P} \mu$, i.e. $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Proof. By Equation 5, $\text{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev gives $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. $\square$

Theorem (Central limit theorem). If X_i are i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \quad (7)$$

The proof goes through characteristic functions and is not reproduced here. The Berry–Esseen theorem quantifies the rate: the Kolmogorov distance to Φ is at most $C E|X_1 - \mu|^3 / (\sigma^3 \sqrt{n})$. This is why the sample size needed depends on skewness.

Theorem (Slutsky and delta method). If $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (constant), then $A_n + B_n \xrightarrow{d} A + b$ and $A_n B_n \xrightarrow{d} bA$. If moreover $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \tau^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, g'(\theta)^2 \tau^2). \quad (8)$$

Slutsky is what licenses replacing an unknown σ by a consistent estimate in a pivot, which is how the Wald and large-sample t intervals are justified.

2 References

- G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed., Duxbury, 2002, chapters 1 to 5.
- J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, Cengage, 2011, chapters 1 to 5.
- L. Wasserman, *All of Statistics*, Springer, 2004, chapters 1 to 5.